Activity

Take two test tubes and label them as A and B. Add 7 ml of 0.1N oxalic acid solution, 5 ml of 0.1N $KMnO_4$ solution and 5 ml of 2N dilute H_2SO_4 in both the test tubes. The colour of the solution is pink in both the test tubes.

Now add few crystals of manganese sulphate to the content in test tube A. the pink colour fades up and disappears. In this case, $MnSO_4$ acts as a catalyst and increases the rate of oxidation of $C_2O_4^{2-}$ by MnO_4^-



Chemical kinetics in pharmaceuticals

Chemical kinetics has many applications in the field of pharmaceuticals. It is used to study the lifetimes and bioavailability of drugs within the body and this branch of study is called **pharmacokinetics** Doctors usually prescribe drugs to be taken at different times of the day. i.e.some drugs are to be taken twice a day, while others are taken three times a day, or just once a day. Pharmacokinetic studies is used to determine the prescription (dosage and frequency). For example, Paracetamol is a well known anti-pyretic and analgesic that is prescribed in cases of fever and body pain. Pharmacokinetic studies showed that Paracetamol has a half-life of 2.5 hours within the body i.e.the plasma concentration of a drug is halved after 2.5 hrs. After 10 hours (4 half-lives)only 6.25 % of drug remains. Based on such studies the dosage and frequency will be decided. In case of paracetamol, it is usually prescribed to take once in 6 hours depending upon the conditions.

Summary

- Chemical kinetics is the study of the rate and the mechanism of chemical reactions, proceeding under given conditions of temperature, pressure, concentration etc.
- The change in the concentration of the species involved in a chemical reaction per unit time gives the rate of a reaction.
- The rate of the reaction, at a particular instant during the reaction is called the instantaneous rate. The shorter the time period, we choose, the closer we approach to the instantaneous rate,
- The rate represents the speed at which the reactants are converted into products at any instant.
- The rate constant is a proportionality constant and It is equal to the rate of reaction, when the concentration of each of the reactants is unity.
- Molecularity of a reaction is the total number of reactant species that are involved in an elementary step.
- The half life of a reaction is defined as the time required for the reactant concentration to reach one half its initial value. For a first order reaction, the half life is a constant

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